

PostgreSQL

```
sudo docker run --name my-net-postgres-db \  
> --network otus-app-network \  
> -e POSTGRES_USER=otus \  
> -e POSTGRES_PASSWORD=myotuspassword \  
> -d \  
> -p 5432:5432 \  
> postgres:latest
```

```
CREATE TABLE sales(  
  
    id integer GENERATED ALWAYS AS IDENTITY NOT NULL,  
  
    create_time date,  
  
    "action" varchar(255),  
  
    PRIMARY KEY(id)  
  
);
```

```
INSERT INTO sales (create_time, "action")  
VALUES  
    ('2024-01-15', 'purchase'),  
    ('2024-02-20', 'refund'),  
    ('2024-03-10', 'purchase'),  
    ('2024-04-05', 'exchange'),  
    ('2024-05-12', 'purchase'),  
    ('2024-06-18', 'cancel'),  
    ('2024-07-22', 'purchase'),  
    ('2024-08-30', 'refund'),  
    ('2024-09-14', 'purchase'),  
    ('2024-10-01', 'exchange'),  
    ('2024-11-07', 'purchase'),  
    ('2024-12-25', 'cancel'),  
    ('2025-01-03', 'purchase'),  
    ('2025-02-14', 'refund'),  
    ('2025-03-21', 'purchase'),  
    ('2025-04-10', 'exchange');
```

```
SELECT action, create_time FROM postgresql('158.160.37.66:5432', 'postgres', 'sales', 'otus', 'myotuspassword');
```

```
SELECT action, create_time FROM postgresql('158.160.37.66:5432', 'postgres', 'sales', 'otus', 'myotuspassword');
```

```
SELECT
    action,
    create_time
FROM postgresql('158.160.37.66:5432', 'postgres', 'sales', 'otus', 'myotuspassword')
```

Query id: a10f7033-e149-4320-9aa6-fc415d86e7d8

	action	create_time
1.	purchase	2024-01-15
2.	refund	2024-02-20
3.	purchase	2024-03-10
4.	exchange	2024-04-05
5.	purchase	2024-05-12
6.	cancel	2024-06-18
7.	purchase	2024-07-22
8.	refund	2024-08-30
9.	purchase	2024-09-14
10.	exchange	2024-10-01
11.	purchase	2024-11-07
12.	cancel	2024-12-25
13.	purchase	2025-01-03
14.	refund	2025-02-14
15.	purchase	2025-03-21
16.	exchange	2025-04-10

16 rows in set. Elapsed: 0.018 sec.

```
CREATE TABLE sales_pg_
(
    id UInt64,
    create_time Date,
    "action" String
)
ENGINE = PostgreSQL('158.160.37.66:5432', 'postgres', 'sales', 'otus', 'myotuspassword');
```

```
select * from sales_pg_
```

```
SELECT *
FROM sales_pg_
```

Query id: 6005dd12-cc93-4329-be65-148220cf00d3

	id	create_time	action
1.	1	2024-01-15	purchase
2.	2	2024-02-20	refund
3.	3	2024-03-10	purchase
4.	4	2024-04-05	exchange
5.	5	2024-05-12	purchase
6.	6	2024-06-18	cancel
7.	7	2024-07-22	purchase
8.	8	2024-08-30	refund

9.	9	2024-09-14	purchase
10.	10	2024-10-01	exchange
11.	11	2024-11-07	purchase
12.	12	2024-12-25	cancel
13.	13	2025-01-03	purchase
14.	14	2025-02-14	refund
15.	15	2025-03-21	purchase
16.	16	2025-04-10	exchange

16 rows in set. Elapsed: 0.017 sec.

Создаем Базу данных в postgresQL

```
CREATE DATABASE otus_clickhouse
```

Внутри выполняем скрипт:

```
DROP TABLE IF EXISTS transaction;
DROP TABLE IF EXISTS sales;
DROP TABLE IF EXISTS product;
DROP TABLE IF EXISTS clients;
```

```
-- 1. Таблица клиентов
CREATE TABLE clients (
    client_id SERIAL PRIMARY KEY,
    client_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    created_at DATE DEFAULT CURRENT_DATE
);
```

```
-- 2. Таблица продуктов
CREATE TABLE product (
    product_id SERIAL PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    price DECIMAL(10, 2) CHECK (price > 0)
);
```

```
-- 3. Таблица продаж (sales)
-- Каждая продажа связана с клиентом и продуктом
CREATE TABLE sales (
    sale_id SERIAL PRIMARY KEY,
    client_id INT NOT NULL REFERENCES clients(client_id) ON DELETE CASCADE,
    product_id INT NOT NULL REFERENCES product(product_id) ON DELETE CASCADE,
    quantity INT CHECK (quantity > 0),
    sale_date DATE DEFAULT CURRENT_DATE
);
```

```
-- 4. Таблица транзакций
-- Каждая транзакция связана с продажей и клиентом
CREATE TABLE transaction (
    transaction_id SERIAL PRIMARY KEY,
    sale_id INT NOT NULL REFERENCES sales(sale_id) ON DELETE CASCADE,
    client_id INT NOT NULL REFERENCES clients(client_id) ON DELETE CASCADE,
    amount DECIMAL(12, 2) NOT NULL,
    transaction_date DATE DEFAULT CURRENT_DATE,
    status VARCHAR(20) CHECK (status IN ('completed', 'pending', 'failed'))
);
```

```
-- Вставка 20+ клиентов
INSERT INTO clients (client_name, email)
SELECT
    'Client ' || g,
    'client' || g || '@example.com'
FROM generate_series(1, 25) AS g;
```

```

-- Вставка 20+ продуктов
INSERT INTO product (product_name, category, price)
VALUES
    ('Laptop', 'Electronics', 999.99),
    ('Smartphone', 'Electronics', 699.99),
    ('Tablet', 'Electronics', 399.99),
    ('Headphones', 'Electronics', 149.99),
    ('Smartwatch', 'Electronics', 249.99),
    ('Coffee Maker', 'Home', 89.99),
    ('Blender', 'Home', 59.99),
    ('Toaster', 'Home', 39.99),
    ('Vacuum Cleaner', 'Home', 199.99),
    ('Microwave', 'Home', 129.99),
    ('Running Shoes', 'Sport', 119.99),
    ('Yoga Mat', 'Sport', 29.99),
    ('Dumbbell Set', 'Sport', 149.99),
    ('Treadmill', 'Sport', 899.99),
    ('Water Bottle', 'Sport', 19.99),
    ('Desk Chair', 'Furniture', 199.99),
    ('Office Desk', 'Furniture', 299.99),
    ('Bookshelf', 'Furniture', 149.99),
    ('Sofa', 'Furniture', 799.99),
    ('Bed Frame', 'Furniture', 499.99),
    ('LED Monitor', 'Electronics', 299.99),
    ('Wireless Mouse', 'Electronics', 29.99),
    ('Keyboard', 'Electronics', 79.99),
    ('Backpack', 'Accessories', 49.99),
    ('Notebook', 'Stationery', 5.99);

-- Вставка 30+ продаж (гарантируем ссылки на существующие client_id и product_id)
INSERT INTO sales (client_id, product_id, quantity, sale_date)
SELECT
    (random() * 24 + 1)::INT + 1, -- client_id от 1 до 25
    (random() * 24 + 1)::INT + 1, -- product_id от 1 до 25
    (random() * 4 + 1)::INT,      -- quantity от 1 до 5
    CURRENT_DATE - (random() * 365)::INT -- дата за последний год
FROM generate_series(1, 30);

-- Вставка 30+ транзакций (связанных с продажами и клиентами)
INSERT INTO transaction (sale_id, client_id, amount, transaction_date, status)
SELECT
    s.sale_id,
    s.client_id,
    s.quantity * p.price * (0.95 + random() * 0.1)::DECIMAL, -- небольшой разброс
    s.sale_date,
    CASE (random() * 2)::INT
        WHEN 0 THEN 'completed'
        WHEN 1 THEN 'pending'
        ELSE 'failed'
    END
FROM sales s
JOIN product p ON s.product_id = p.product_id
LIMIT 30;

-- Проверка количества строк
SELECT
    (SELECT COUNT(*) FROM clients) AS clients_count,
    (SELECT COUNT(*) FROM product) AS product_count,
    (SELECT COUNT(*) FROM sales) AS sales_count,
    (SELECT COUNT(*) FROM transaction) AS transaction_count;

```

В файле /var/lib/postgresql/18/docker/postgresql.conf
Добавляем :

```
#max_wal_senders = 10          # max number of walsender processes
wal_level=logical              # (change requires restart)
max_replication_slots = 10     # max number of replication slots
                                # (change requires restart)
#wal_keep_size = 0             # in megabytes; 0 disables
#max_slot_wal_keep_size = -1   # in megabytes; -1 disables
#idle_replication_slot_timeout = 0 # in seconds; 0 disables
#wal_sender_timeout = 60s      # in milliseconds; 0 disables
#track_commit_timestamp = off  # collect timestamp of transaction commit
                                # (change requires restart)

# - Primary Server -

# These settings are ignored on a standby server.
```

Создаем Базу данных с синхронизацией из postgresQL

```
SET allow_experimental_database_materialized_postgresql=1
```

```
SET allow_experimental_database_materialized_postgresql = 1
```

Query id: aaed1bc0-2a81-4780-b1b1-37a82ff66211

Ok.

0 rows in set. Elapsed: 0.001 sec.

```
clickhouse-node.ru-central1.internal :) CREATE DATABASE postgres_db_otus_
ENGINE = MaterializedPostgreSQL('158.160.37.66:5432', 'otus_clickhouse', 'otus', 'myotuspassword')
;
```

```
CREATE DATABASE postgres_db_otus_
ENGINE = MaterializedPostgreSQL('158.160.37.66:5432', 'otus_clickhouse', 'otus', 'myotuspassword')
```

Query id: 3d4fca8a-0498-4d71-9015-69ed770e75be

Ok.

0 rows in set. Elapsed: 0.019 sec.

```
select * from postgres_db_otus_.clients
```

```
SELECT *
FROM postgres_db_otus_.clients
```

Query id: f9c45af9-85c8-4f9a-87c9-fd23226c9fe6

	client_id	client_name	email	created_at
1.	1	Client 1	client1@example.com	2025-10-28
2.	2	Client 2	client2@example.com	2025-10-28
3.	3	Client 3	client3@example.com	2025-10-28
4.	4	Client 4	client4@example.com	2025-10-28
5.	5	Client 5	client5@example.com	2025-10-28
6.	6	Client 6	client6@example.com	2025-10-28
7.	7	Client 7	client7@example.com	2025-10-28
8.	8	Client 8	client8@example.com	2025-10-28
9.	9	Client 9	client9@example.com	2025-10-28
10.	10	Client 10	client10@example.com	2025-10-28
11.	11	Client 11	client11@example.com	2025-10-28
12.	12	Client 12	client12@example.com	2025-10-28
13.	13	Client 13	client13@example.com	2025-10-28
14.	14	Client 14	client14@example.com	2025-10-28
15.	15	Client 15	client15@example.com	2025-10-28
16.	16	Client 16	client16@example.com	2025-10-28
17.	17	Client 17	client17@example.com	2025-10-28
18.	18	Client 18	client18@example.com	2025-10-28
19.	19	Client 19	client19@example.com	2025-10-28

20.	20	Client 20	client20@example.com	2025-10-28
21.	21	Client 21	client21@example.com	2025-10-28
22.	22	Client 22	client22@example.com	2025-10-28
23.	23	Client 23	client23@example.com	2025-10-28
24.	24	Client 24	client24@example.com	2025-10-28
25.	25	Client 25	client25@example.com	2025-10-28

25 rows in set. Elapsed: 0.003 sec.

Заголовок